

3	(a)	The graph shows a <u>linear line passing through the origin</u>. This shows that acceleration is proportional to displacement. The <u>graph shows a negative gradient</u>. This means that acceleration and displacement is always in opposite direction.	B1 B1
	(b)	$a = -\omega^2 x$ Hence, using one of the data point $(-0.04, 0.32)$, $0.32 = -\omega^2(-0.04)$ $\omega^2 = 8.0$ $\omega = \sqrt{\frac{k}{L}} \Rightarrow k = \omega^2 L$ $k = (8.0)(1.24)$ $= 9.92 \text{ m s}^{-2}$	C1 C1 A1
	(c)	<u>Increasing the length decreases the angular frequency.</u> Since total energy of oscillation $= \frac{1}{2} m \omega^2 x_0^2$, and total energy does not change, the amplitude is inversely proportional to angular frequency². <u>increasing the length will increase the amplitude</u> of oscillation.	M1 A1

4	(a)	Coulomb's law states that the magnitude of the electric force between two point	B1
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